Dyne Target Literature

April 2020



MAY 4

Dyne Therapeutics

Dyne THERAPEUTICS

DM1 Journal Articles

April 1 – April 30, 2020

Myotonic has been generating a monthly research newsletter with a running list of current journal publications for several years. *It ended in December 2019*, but I recommend reviewing the list here for a good overview of the literature on myotonic dystrophy:
https://www.myotonic.org/community/news/Research

Ventral tegmental area dysfunction affects decision-making in patients with myotonic dystrophy type 1
Serra L, Scocchia M, Meola G, D'Amelio M, Bruschini M, Silvestri G, Petrucci A, Di Domenico C, Caltagirone C, Koch G, Cercignani M, Petrosini L, Bozzali M. Cortex. 2020 Apr 8;128:192-202. doi: 10.1016/j.cortex.2020.03.022. Online ahead of print. PMID: 32361267
The clinical manifestations of Myotonic Dystrophy type-1 (DM1) are associated with a complex mixture of multisystem features including cognitive dysfunctions that strongly impact on patients' social and occupational functioning. ...

MyomiRNAs and myostatin as physical rehabilitation biomarkers for myotonic dystrophy.
Pegoraro V, Cudia P, Baba A, Angelini C. Neurol Sci. 2020 Apr 29. doi: 10.1007/s10072-020-04409-2. Online ahead of print. PMID: 32350671
MiR-1 and myostatin are markers for muscle growth and regeneration. Myostatin has a key role in the regulation of muscle mass. Myotonic dystrophy type 1(DM1) patients have a disease-specific serum miRNA profile characterized by upregulation of miR-1, miR-206, miR-133a, and miR-133b (myomiRNAs).This study aims to evaluate the possible utility of myomiRs and myostatin as biomarkers of rehabilitation efficacy in DM1, supporting clinical outcomes that are often variable and related to the patient's clinical condition. In 9 genetically proven DM1 patients, we collected biological samples before (T0) and after (T1) exercise rehabilitation training as biological measurement. ...

CDK12 inhibition reduces abnormalities in cells from patients with myotonic dystrophy and in a mouse model.
Ketley A, Wojciechowska M, Ghidelli-Disse S, Bamborough P, Ghosh TK, Morato ML, Sedehizadeh S, Malik NA, Tang Z, Powalowska P, Tanner M, Billeter-Clark R, Trueman RC, Geiszler PC, Agostini A, Othman O, Bösche M, Bantscheff M, Rüdiger M, Mossakowska DE, Drewry DH, Zuercher WJ, Thornton CA, Drewes G, Uings I, Hayes CJ, Brook JD. Sci Transl Med. 2020 Apr 29;12(541):eaaz2415. doi: 10.1126/scitranslmed.aaz2415. PMID: 32350131

Myotonic dystrophy type 1 (DM1) is an RNA-based disease with no current treatment. It is caused by a transcribed CTG repeat expansion within the 3' untranslated region of the dystrophia myotonica protein kinase (DMPK) gene. ...

ExpansionHunter Denovo: a computational method for locating known and novel repeat expansions in short-read sequencing data.

Dolzhenko E, Bennett MF, Richmond PA, Trost B, Chen S, van Vugt JJFA, Nguyen C, Narzisi G, Gainullin VG, Gross AM, Lajoie BR, Taft RJ, Wasserman WW, Scherer SW, Veldink JH, Bentley DR, Yuen RKC, Bahlo M, Eberle MA. Genome Biol. 2020 Apr 28;21(1):102. doi: 10.1186/s13059-020-02017-z.PMID: 32345345 Free PMC article.

Strength-training effectively alleviates skeletal muscle impairments in myotonic dystrophy type 1.

Roussel MP, Hébert LJ, Duchesne E. Neuromuscul Disord. 2020 Feb 28:S0960-8966(20)30055-9. doi: 10.1016/j.nmd.2020.02.015. Online ahead of print. PMID: 32340814

Myotonic dystrophy type 1 (DM1) is a multisystemic disease characterized by progressive muscle weakness. The aim of this project is to evaluate the effects of a 12-week lower limb strength training program in 11 men with DM1. ...

Differential diagnosis of temporal lobe lesions with hyperintense signal on T2-weighted and FLAIR sequences: pictorial essay.

Santana LM, Valadares EJA, Rosa-Júnior M. Radiol Bras. 2020 Mar-Apr;53(2):129-136. doi: 10.1590/0100-3984.2018.0117.PMID: 32336830 Free PMC article.

This pictorial essay aims to illustrate the relatively common diseases that affect the temporal lobe, such as herpes simplex encephalitis, neurosyphilis, limbic encephalitis, postictal edema, neoplasia, and multiple sclerosis, as well as those that are less common, such as myotonic dystrophy type 1, CADASIL, and CARASIL, together with the particularities of each entity....

Loss of MBNL1 induces RNA misprocessing in the thymus and peripheral blood.

Sznajder ŁJ, Scotti MM, Shin J, Taylor K, Ivankovic F, Nutter CA, Aslam FN, Subramony SH, Ranum LPW, Swanson MS. Nat Commun. 2020 Apr 24;11(1):2022. doi: 10.1038/s41467-020-15962-x.PMID: 32332745 Free PMC article.

Here we demonstrate that the RNA alternative splicing factor MBNL1, which is sequestered in nuclear RNA foci by C(C)UG microsatellite expansions in myotonic dystrophy (DM), is essential for normal thymus development and function. ...CNBP, the gene containing an intronic CCTG microsatellite expansion in DM type 2 (DM2), is coordinately expressed with MBNL1 in the developing thymus and DM2 CCTG expansions induce similar transcriptome alterations in DM2 blood, which thus serve as disease-specific biomarkers....

Validity of the Mini-BESTest in adults with myotonic dystrophy type 1.

Duchesne E, Hébert LJ, Mathieu J, Côté I, Roussel MP, Gagnon C. Muscle Nerve. 2020 Apr 20. doi: 10.1002/mus.26893. Online ahead of print. PMID: 32314404

INTRODUCTION: Myotonic dystrophy type 1 (DM1) is a multisystemic neuromuscular disease that causes balance problems. The objective of this study was to assess the construct validity of the Mini-BESTest among adults with DM1. ...

Myotonic Dystrophy type 1 cells display impaired metabolism and mitochondrial dysfunction that are reversed by metformin.

García-Puga M, Saenz-Antoñanzas A, Fernández-Torrón R, Munain AL, Matheu A. Aging (Albany NY). 2020 Apr 8;12(7):6260-6275. doi: 10.18632/aging.103022. Epub 2020 Apr 8. PMID: 32310829 Free PMC article.

Myotonic dystrophy type 1 (DM1; MIM #160900) is an autosomal dominant disorder, clinically characterized by progressive muscular weakness and multisystem degeneration. ...

Screening for early symptoms of respiratory involvement in myotonic dystrophy type 1 using the Respicheck questionnaire.

De Mattia E, Lizio A, Falcier E, Sannicolò G, Gualandris M, Rossi G, Zanolini A, Pozzi S, Messina S, Sframeli M, Lunetta C, Rao F, Sansone VA. Neuromuscul Disord. 2020 Feb 29:S0960-8966(20)30054-7. doi: 10.1016/j.nmd.2020.02.014. Online ahead of print. PMID: 32305258

Symptoms of respiratory involvement are frequently present but overlooked by patients with Myotonic Dystrophy type 1 (DM1). A respiratory symptom checklist was designed to test whether a DM-specifically designed checklist to detect symptoms of respiratory involvement (The Respicheck Questionnaire) could help patients be more aware of their respiratory problems, if any, and help clinicians in identifying potential candidates for intervention. ...

Swallow-related quality of life and oropharyngeal dysphagia in myotonic dystrophy.

Pilz W, Passos VL, Verdonschot RJ, Meijers J, Roodenburg N, Halmans Y, Faber CG, Kremer B, Baijens LWJ. Eur Arch Otorhinolaryngol. 2020 Apr 15. doi: 10.1007/s00405-020-05964-2. Online ahead of print. PMID: 32296979

PURPOSE: This study describes swallow-related quality of life (SWAL-QOL) in patients with myotonic dystrophy type 1 (DM1) and investigates its association with swallowing function and disease severity.

Swallow-related quality of life and oropharyngeal dysphagia in myotonic dystrophy.

Pilz W, Passos VL, Verdonschot RJ, Meijers J, Roodenburg N, Halmans Y, Faber CG, Kremer B, Baijens LWJ. Eur Arch Otorhinolaryngol. 2020 Apr 15. doi: 10.1007/s00405-020-05964-2. Online ahead of print. PMID: 32296979

PURPOSE: This study describes swallow-related quality of life (SWAL-QOL) in patients with myotonic dystrophy type 1 (DM1) and investigates its association with swallowing function and disease severity. ...

Towards development of a statistical framework to evaluate myotonic dystrophy type 1 mRNA biomarkers in the context of a clinical trial.

Kurkiewicz A, Cooper A, McIlwaine E, Cumming SA, Adam B, Krahe R, Puymirat J, Schoser B, Timchenko L, Ashizawa T, Thornton CA, Rogers S, McClure JD, Monckton DG. PLoS One. 2020 Apr 14;15(4):e0231000. doi: 10.1371/journal.pone.0231000. eCollection 2020. PMID: 32287265 Free PMC article.

Myotonic dystrophy type 1 (DM1) is a rare genetic disorder, characterised by muscular dystrophy, myotonia, and other symptoms. ...

Rare Disease: Cardiac Risk Assessment With MRI in Patients With Myotonic Dystrophy Type 1.

Alì M, Monti CB, Melazzini L, Cardani R, Fossati B, Cavalli M, Chow K, Secchi F, Meola G, Sardanelli F. Front Neurol. 2020 Mar 19;11:192. doi: 10.3389/fneur.2020.00192. eCollection 2020.PMID: 32265828 Free PMC article.

Introduction: To evaluate myocardial strain and extracellular volume in myotonic dystrophy type 1 (DM1) patients as potential imaging biomarkers of subclinical cardiac pathology. ...Results: Global cardiac strain (median -19.1%; IQR -20.5%, -16.5%) in DM1 patients was lower (p = 0.011) than that in controls (median-21.7%; IQR-22.7%,-21.3%). ...

Systemic Therapy in a RNA Toxicity Mouse Model with an Antisense Oligonucleotide Therapy Targeting a non-CUG sequence within the DMPK 3'UTR RNA.

Yadava RS, Yu Q, Mandal M, Rigo F, Bennett CF, Mahadevan MS. Hum Mol Genet. 2020 Apr 2:ddaa060. doi: 10.1093/hmg/ddaa060. Online ahead of print. PMID: 32242217

Myotonic dystrophy type 1 (DM1), the most common adult muscular dystrophy, is an autosomal dominant disorder caused by an expansion of a (CTG)n tract within the 3' untranslated region (3'UTR) of the DMPK (dystrophia myotonica protein kinase) gene. ...

DMD Journal Articles

April 1 – April 30, 2020

In vitro assay for the efficacy assessment of AAV vectors expressing microdystrophin.

Danilov KA, Vassilieva SG, Polikarpova AV, Starikova AV, Shmidt AA, Galkin II, Tsitrina AA, Egorova TV, Orlov SN, Kotelevtsev YV. Exp Cell Res. 2020 Apr 29:112033. doi: 10.1016/j.yexcr.2020.112033. Online ahead of print. PMID: 32360435
AAV-delivered microdystrophin genes hold great promise for Duchenne muscular dystrophy (DMD) treatment. It is anticipated that the optimization of engineered dystrophin genes will be required to increase the efficacy and reduce the immunogenicity of transgenic proteins. ...

Cognitive impairment appears progressive in the mdx mouse.
Bagdatlioglu E, Porcari P, Greally E, Blamire AM, Straub VW. Neuromuscul Disord. 2020 Mar 4:S0960-8966(20)30058-4. doi: 10.1016/j.nmd.2020.02.018. Online ahead of print. PMID: 32360405
Duchenne muscular dystrophy (DMD) is an X-linked recessive muscle wasting disease caused by mutations in the DMD gene, which encodes the large cytoskeletal protein dystrophin. ...

Acute conversion of patient-derived Duchenne muscular dystrophy iPSC into myotubes reveals constitutive and inducible over-activation of TGFβ-dependent pro-fibrotic signaling.
Caputo L, Granados A, Lenzi J, Rosa A, Ait-Si-Ali S, Puri PL, Albini S. Skelet Muscle. 2020 May 2;10(1):13. doi: 10.1186/s13395-020-00224-7.PMID: 32359374
BACKGROUND: In Duchenne muscular dystrophy (DMD), DYSTROPHIN deficiency exposes myofibers to repeated cycles of contraction/degeneration, ultimately leading to muscle loss and replacement by fibrotic tissue. ...

Mutation Spectrum of Dystrophinopathies in India: Implications for Therapy.
Kohli S, Saxena R, Thomas E, Singh K, Bijarnia Mahay S, Puri RD, Verma IC. Indian J Pediatr. 2020 May 2. doi: 10.1007/s12098-020-03286-z. Online ahead of print. PMID: 32358784
METHODS: In 1660 patients diagnosed as Duchenne muscular dystrophy (DMD) /Becker muscular dystrophy (BMD) deletion- duplication analysis of all 79 exons was carried out using Multiplex ligation-dependent probe amplification (MLPA) technology. ...

Characterization of Forearm Muscle Activation in Duchenne Muscular Dystrophy via High-Density Electromyography: A Case Study on the Implications for Myoelectric Control.
Nizamis K, Rijken NHM, van Middelaar R, Neto J, Koopman BFJM, Sartori M. Front Neurol. 2020 Apr 15;11:231. doi: 10.3389/fneur.2020.00231. eCollection 2020.PMID: 32351441 Free PMC article.
Duchenne muscular dystrophy (DMD) is a genetic disorder that results in progressive muscular degeneration. Although medical advances increased their life expectancy, DMD individuals are still highly dependent on caregivers. ...Traditional low-density sEMG was used in the past to explore the muscular activations of individuals with DMD; however, it cannot provide high spatial resolution. ...

Noncoding RNAs in Duchenne and Becker muscular dystrophies: role in pathogenesis and future prognostic and therapeutic perspectives.

Brusa R, Magri F, Bresolin N, Comi GP, Corti S. Cell Mol Life Sci. 2020 Apr 29. doi: 10.1007/s00018-020-03537-4. Online ahead of print. PMID: 32350552 Review.

Noncoding RNAs (ncRNAs), such as miRNAs and long noncoding RNAs, are key regulators of gene expression at the post-transcriptional level and represent promising therapeutic targets and biomarkers for several human diseases, including Duchenne and Becker muscular dystrophies (DMD/BMD). A role for ncRNAs in the pathogenesis of muscular dystrophies has been suggested, even if it is still incompletely understood. ...

Generation of the Becker muscular dystrophy patient derived induced pluripotent stem cell line carrying the DMD splicing mutation c.1705-8 T>C.

Rovina D, Castiglioni E, Niro F, Farini A, Belicchi M, Di Fede E, Gervasini C, Paganini S, Di Segni M, Torrente Y, Santoro R, Pompilio G, Gowran A. Stem Cell Res. 2020 Apr 22;45:101819. doi: 10.1016/j.scr.2020.101819. Online ahead of print. PMID: 32348941

Becker Muscular dystrophy (BMD) is an X-linked syndrome characterized by progressive muscle weakness. BMD is generally less severe than Duchenne Muscular Dystrophy. ...

Combined Treatment with Peptide-Conjugated Phosphorodiamidate Morpholino Oligomer-PPMO and AAV-U7 Rescues the Severe DMD Phenotype in Mice.

Forand A, Muchir A, Mougenot N, Sevoz-Couche C, Peccate C, Lemaitre M, Izabelle C, Wood M, Lorain S, Piétri-Rouxel F. Mol Ther Methods Clin Dev. 2020 Mar 17;17:695-708. doi: 10.1016/j.omtm.2020.03.011. eCollection 2020 Jun 12. PMID: 32346547 Free PMC article.

Duchenne muscular dystrophy (DMD) is a devastating neuromuscular disease caused by an absence of the dystrophin protein, which is essential for muscle fiber integrity. ...

Duchenne muscular dystrophy hiPSC-derived myoblast drug screen identifies compounds that ameliorate disease in mdx mice.

Sun C, Choi IY, Rovira Gonzalez YI, Andersen P, Talbot CC Jr, Iyer SR, Lovering RM, Wagner KR, Lee G. JCI Insight. 2020 Apr 28:134287. doi: 10.1172/jci.insight.134287. Online ahead of print. PMID: 32343677 Free article.

Duchenne muscular dystrophy (DMD) is the most common muscular dystrophy. When human induced pluripotent stem cells (hiPSCs) were differentiated into myoblasts, the myoblasts derived from DMD patients' hiPSCs (DMD hiPSC-derived myoblasts) exhibited an identifiable DMD relevant phenotype: myogenic fusion deficiency. ..

Genetic modifiers of respiratory function in Duchenne muscular dystrophy.

Bello L, D'Angelo G, Villa M, Fusto A, Vianello S, Merlo B, Sabbatini D, Barp A, Gandossini S, Magri F, Comi GP, Pedemonte M, Tacchetti P, Lanzillotta V, Trucco F, D'Amico A, Bertini E, Astrea G, Politano L,

Masson R, Baranello G, Albamonte E, De Mattia E, Rao F, Sansone VA, Previtali S, Messina S, Vita GL, Berardinelli A, Mongini T, Pini A, Pane M, Mercuri E, Vianello A, Bruno C, Hoffman EP, Morgenroth L, Gordish-Dressman H, McDonald CM; CINRG-DNHS Investigators, Pegoraro E. Ann Clin Transl Neurol. 2020 Apr 28. doi: 10.1002/acn3.51046. Online ahead of print. PMID: 32343055 Free article.
OBJECTIVE: Respiratory insufficiency is a major complication
of Duchenne muscular dystrophy (DMD). Its progression shows considerable interindividual variability, which has been less thoroughly characterized and understood than in skeletal muscle. ...Genetic association findings were independently validated in the Cooperative International Neuromuscular Research Group Duchenne Natural History Study (CINRG-DNHS). ...

Effect of Ataluren on dystrophin mutations.

Berger J, Li M, Berger S, Meilak M, Rientjes J, Currie PD.J Cell Mol Med. 2020 Apr 28. doi: 10.1111/jcmm.15319. Online ahead of print. PMID: 32343037 Free article.
Duchenne muscular dystrophy is a severe muscle wasting disease caused by mutations in the dystrophin gene (dmd). Ataluren has been approved by the European Medicines Agency for treatment of Duchenne muscular dystrophy. ...

Heterogenetic parabiosis between healthy and dystrophic mice improve the histopathology in muscular dystrophy.

Lu A, Guo P, Wang L, Tseng C, Huard M, Allen C, McCarrick-Walmsley R, Whitney KE, Huard J.Sci Rep. 2020 Apr 27;10(1):7075. doi: 10.1038/s41598-020-64042-z.PMID: 32341395 Free PMC article.
Duchenne muscular dystrophy (DMD) is a progressive muscle disease, characterized by mutations in the X-linked dystrophin, that has several therapeutic options but no curative treatment. ...

Clinical utility of multigene analysis in over 25,000 patients with neuromuscular disorders.

Winder TL, Tan CA, Klemm S, White H, Westbrook JM, Wang JZ, Entezam A, Truty R, Nussbaum RL, McNally EM, Aradhya S. Neurol Genet. 2020 Mar 9;6(2):e412. doi: 10.1212/NXG.0000000000000412. eCollection 2020 Apr. PMID: 32337338 Free PMC article.
RESULTS: A definitive molecular diagnosis was obtained in 20% of this cohort, with yields ranging from 4% among individuals with congenital myasthenic syndrome to 33% among those with
a muscular dystrophy. ...Even for classic disorders like Duchenne muscular dystrophy, at least 49% of clinically significant results were identified through gene panels intended for differential diagnoses rather than through single-gene analysis. ...

Identification of therapeutics that target eEF1A2 and upregulate utrophin A translation in dystrophic muscles.

Péladeau C, Adam N, Bronicki LM, Coriati A, Thabet M, Al-Rewashdy H, Vanstone J, Mears A, Renaud JM, Holcik M, Jasmin BJ. Nat Commun. 2020 Apr 24;11(1):1990. doi: 10.1038/s41467-020-15971-w.PMID: 32332749 Free PMC article.

Up-regulation of utrophin in muscles represents a promising therapeutic strategy for the treatment of Duchenne Muscular Dystrophy. ...

Risk factors associated with prolonged intensive care unit stay after posterior spinal fusion for Duchenne muscular dystrophy.

Saito W, Inoue G, Shirasawa E, Imura T, Nakazawa T, Miyagi M, Uchida K, Matsuda H, Akazawa T, Takaso M. J Orthop Sci. 2020 Apr 21:S0949-2658(20)30085-3. doi: 10.1016/j.jos.2020.03.014. Online ahead of print. PMID: 32331988

BACKGROUND: The perioperative complication rate for spinal fusion in Duchenne muscular dystrophy (DMD) remains high and sometimes prolonged perioperative intensive care is needed. ...

Steroid Responsiveness in Duchenne Muscular Dystrophy - Predictive Value of Epigenetic Regulator Histone Deacetylase 2.

Pradhan S, Das A, Singh H, Chaturvedi S, Singh K, Mishra R, Misra DP, Agarwal V. Indian J Pediatr. 2020 Apr 20. doi: 10.1007/s12098-020-03183-5. Online ahead of print. PMID: 32314166

OBJECTIVE: To find the role of an epigenetic regulator histone deacetylase (HDAC-2) in predicting steroid responsiveness in patients with Duchenne muscular dystrophy (DMD). ...

Prostaglandin EP2 receptor downstream of Notch signaling inhibits differentiation of human skeletal muscle progenitors in differentiation conditions.

Sakai-Takemura F, Nogami K, Elhussieny A, Kawabata K, Maruyama Y, Hashimoto N, Takeda S, Miyagoe-Suzuki Y. Commun Biol. 2020 Apr 20;3(1):182. doi: 10.1038/s42003-020-0904-6.PMID: 32313117 Free PMC article.

Understanding the signaling pathways that regulate proliferation and differentiation of muscle progenitors is essential for successful cell transplantation for treatment of Duchenne muscular dystrophy. ...

Assessing the effect of non-invasive ventilation on cardiac function in Duchenne muscular dystrophy **requires prospective studies.**

Finsterer J, Stollberger C. Cardiol Young. 2020 Apr 21:1-2. doi: 10.1017/S1047951120000839. Online ahead of print. PMID: 32312346 No abstract available.

Myocyte-derived Myomaker expression is required for regenerative fusion but exacerbates membrane instability in dystrophic myofibers.

Petrany MJ, Song T, Sadayappan S, Millay DP.JCI Insight. 2020 Apr 14:136095. doi: 10.1172/jci.insight.136095. Online ahead of print. PMID: 32310830 Free article.

Chronic muscle regeneration in Duchenne muscular dystrophy (DMD) is characterized by ongoing fusion of satellite cell (SC) progeny, but the effects of fusion on disease and the mechanisms by which fusion is accomplished in this setting are not fully understood. ...

Correlation of Bone Mineral Density with Pulmonary Function in Advanced Duchenne Muscular Dystrophy.

Lee JW, Cho HE, Kang SW, Choi WA, Suh MR, Kim B.PM R. 2020 Apr 18. doi: 10.1002/pmrj.12386. Online ahead of print. PMID: 32306557

However, there is lack of research investigating the correlation between BMD and respiratory function, one of few measurable physical parameters in advanced Duchenne muscular dystrophy (DMD) patients. ...

T_2 relaxation-time mapping in healthy and diseased skeletal muscle using extended phase graph algorithms.

Keene KR, Beenakker JM, Hooijmans MT, Naarding KJ, Niks EH, Otto LAM, van der Pol WL, Tannemaat MR, Kan HE, Froeling M. Magn Reson Med. 2020 Apr 19. doi: 10.1002/mrm.28290. Online ahead of print. PMID: 32306450

A high-throughput digital script for multiplexed immunofluorescent analysis and quantification of sarcolemmal and sarcomeric proteins in muscular dystrophies.

Scaglioni D, Ellis M, Catapano F, Torelli S, Chambers D, Feng L, Sewry C, Morgan J, Muntoni F, Phadke R. Acta Neuropathol Commun. 2020 Apr 17;8(1):53. doi: 10.1186/s40478-020-00918-5.PMID: 32303261 Free PMC article.

The primary molecular endpoint for many Duchenne muscular dystrophy (DMD) clinical trials is the induction, or increase in production, of dystrophin protein in striated muscle. ...In this work, we present further development and optimisation of an automated, digital, high-throughput script for quantitative analysis of multiplexed immunofluorescent (IF) whole slide images (WSI) of dystrophin, dystrophin associated proteins (DAPs) and regenerating myofibres (fetal/developmental myosin-positive) in transverse sections of DMD, Becker muscular dystrophy (BMD) and control skeletal muscle biopsies. ...

Targeted Treatments for Inherited Neuromuscular Diseases of Childhood.

Fay AJ, Knox R, Neil EE, Strober J. Semin Neurol. 2020 Apr 15. doi: 10.1055/s-0040-1702940. Online ahead of print. PMID: 32294764

At the forefront of these advances are the two recently approved treatments for spinal muscular atrophy: one, an antisense oligonucleotide that modifies splicing of the SMN2 gene, and, the other, a gene therapy vector that delivers the SMN1 gene to motor neurons, both of which are allowing patients to acquire developmental milestones previously unseen in this fatal disease. This review highlights these advances and emerging targeted therapies

for Duchenne muscular dystrophy and centronuclear myopathy, while also covering enzyme replacement therapy and small molecule-based targeted therapies for conditions such as Pompe's disease and congenital myasthenic syndromes. ...

Isoform-Specific Na,K-ATPase and Membrane Cholesterol Remodeling in the Motor Endplates in Distinct Mouse Models of Myodystrophy.

Kravtsova VV, Bouzinova EV, Chibalin AV, Matchkov VV, Krivoi II. Am J Physiol Cell Physiol. 2020 Apr 15. doi: 10.1152/ajpcell.00453.2019. Online ahead of print. PMID: 32293933

Mdx and Bla/J mice are the experimental models of Duchenne muscular dystrophy and dysferlinopathy. The molecular mechanism behind myodystrophy is of therapeutic importance; however, the detailed role of Na,K-ATPase in these dysfunctions has not previously been addressed. ...

Mobile arm supports in Duchenne muscular dystrophy: a pilot study of user experience and outcomes.

Cruz A, Callaway L, Randall M, Ryan M. Disabil Rehabil Assist Technol. 2020 Apr 15:1-10. doi: 10.1080/17483107.2020.1749892. Online ahead of print. PMID: 32293211

Purpose: This pilot study examined whether two different types of non-powered mobile arm supports (MAS) enhanced upper limb function and independence with activities of daily living (ADLs) in people with Duchenne muscular dystrophy (DMD).Method: A mixed methods cross-sectional design was used. ...

Beyond ambulation: Measuring physical activity in youth with Duchenne muscular dystrophy.

Killian M, Buchowski MS, Donnelly T, Burnette WB, Markham LW, Slaughter JC, Xu M, Crum K, Damon BM, Soslow JH. Neuromuscul Disord. 2020 Feb 20:S0960-8966(20)30033-X. doi: 10.1016/j.nmd.2020.02.007. Online ahead of print. PMID: 32291149

Patients with Duchenne muscular dystrophy (DMD) develop skeletal muscle weakness and cardiomyopathy. Validated skeletal muscle outcome measures are limited to ambulatory patients, but most DMD patients in cardiac trials are non-ambulatory. ...

Upper and Lower Extremities in Duchenne Muscular Dystrophy Evaluated with Quantitative MRI and Proton MR Spectroscopy in a Multicenter Cohort.

Forbes SC, Arora H, Willcocks RJ, Triplett WT, Rooney WD, Barnard AM, Alabasi U, Wang DJ, Lott DJ, Senesac CR, Harrington AT, Finanger EL, Tennekoon GI, Brandsema J, Daniels MJ, Sweeney HL, Walter GA, Vandenborne K. Radiology. 2020 Apr 14:192210. doi: 10.1148/radiol.2020192210. Online ahead of print. PMID: 32286193

Background Upper extremity MRI and proton MR spectroscopy are increasingly considered to be outcome measures in Duchenne muscular dystrophy (DMD) clinical trials. ...Conclusion This multicenter study demonstrated early and progressive involvement of upper extremity muscles in Duchenne muscular dystrophy (DMD) and showed the feasibility of MRI and (1)H MR spectroscopy

to track disease progression over a wide range of ages in participants with DMD. © RSNA, 2020 Online supplemental material is available for this article....

Effective restoration of dystrophin expression in iPSC ^{Mdx}-derived muscle progenitor cells using the CRISPR/Cas9 system and homology-directed repair technology.

Jin Y, Shen Y, Su X, Weintraub NL, Tang Y. Comput Struct Biotechnol J. 2020 Mar 25;18:765-773. doi: 10.1016/j.csbj.2020.03.012. eCollection 2020.PMID: 32280431 Free PMC article.

Duchenne muscular dystrophy (DMD) is a progressive myopathic disease caused by mutations in the gene encoding dystrophin protein that eventually leads to the exhaustion of myogenic progenitor cells (MPC). ...

Duchenne Muscular Dystrophy: the Heart of the Matter.

Shih JA, Folch A, Wong BL. Curr Heart Fail Rep. 2020 Apr 8. doi: 10.1007/s11897-020-00456-0. Online ahead of print. PMID: 32270339 Review.

PURPOSE OF REVIEW: Duchenne muscular dystrophy is one of many neuromuscular disorders, but it frequently causes severe disability early in life and early death. ...This article reviews the diagnosis and treatment of heart disease in Duchenne muscular dystrophy as well as emerging therapies....

Autophagy: an essential but limited cellular process for timely skeletal muscle recovery from injury.

Call JA, Nichenko AS. Autophagy. 2020 Apr 17:1-4. doi: 10.1080/15548627.2020.1753000. Online ahead of print. PMID: 32267791

The identification of the molecular mechanisms governing autophagy and autophagy flux may serve as targets for future therapies to enhance the recovery of its function in healthy and diseased muscle.

Abbreviations: BNIP3: BCL2/adenovirus E1B interacting protein 3; CQ: chloroquine;

DMD: Duchenne muscular dystrophy; MAP1LC3/LC3: microtubule-associated protein 1 light chain 3; ULK1: unc-51 like kinase 1....

Nutraceutical and pharmaceutical cocktails did not preserve diaphragm muscle function or reduce muscle damage in D2-mdx mice.

Spaulding HR, Quindry T, Quindry JC, Selsby JT. Exp Physiol. 2020 Apr 8. doi: 10.1113/EP087887. Online ahead of print. PMID: 32267561

ABSTRACT: Duchenne muscular dystrophy (DMD) is characterized by the absence of dystrophin protein and causes muscle weakness and muscle injury culminating in respiratory failure and cardiomyopathy...

High-Capacity Adenoviral Vectors Permit Robust and Versatile Testing of *DMD* Gene Repair Tools and Strategies in Human Cells.

Brescia M, Janssen JM, Liu J, Gonçalves MAFV. Cells. 2020 Apr 2;9(4):E869. doi: 10.3390/cells9040869.PMID: 32252479 Free article.

Duchenne muscular dystrophy (DMD) is a fatal X-linked muscle wasting disorder arising from mutations in the ~2.4 Mb dystrophin-encoding DMD gene. ...

Bone health in pediatric patients with neurological disorders.

Ko A, Kong J, Samadov F, Mukhamedov A, Kim YM, Lee YJ, Nam SO. Ann Pediatr Endocrinol Metab. 2020 Mar;25(1):15-23. doi: 10.6065/apem.2020.25.1.15. Epub 2020 Mar 31. PMID: 32252212 Free PMC article.

In this article, mechanisms involved in altered bone health in children with neurological disorders and management for patients with epilepsy, cerebral palsy, and Duchenne muscular dystrophy regarding bone health are reviewed....

TRPCs: Influential Mediators in Skeletal Muscle.

Choi JH, Jeong SY, Oh MR, Allen PD, Lee EH. Cells. 2020 Apr 1;9(4):E850. doi: 10.3390/cells9040850. PMID: 32244622 Free article. Review.

Here, studies on the roles of channel proteins that mediate extracellular Ca(2+) entry into skeletal muscle cells using skeletal myoblasts, myotubes, fibers, tissue, or skeletal muscle-originated cell lines are reviewed with special attention to the proposed functions of transient receptor potential canonical proteins (TRPCs) as store-operated Ca(2+) entry (SOCE) channels under normal conditions and the potential abnormal properties of TRPCs in muscle diseases such as Duchenne muscular dystrophy (DMD)....

Common therapeutic advances for Duchenne muscular dystrophy (DMD).

Salmaninejad A, Jafari Abarghan Y, Bozorg Qomi S, Bayat H, Yousefi M, Azhdari S, Talebi S, Mojarrad M. Int J Neurosci. 2020 Apr 3:1-20. doi: 10.1080/00207454.2020.1740218. Online ahead of print. PMID: 32241218

Background and purpose: Duchenne muscular dystrophy (DMD), a lethal X-linked recessive muscle dystrophy, is resulted in by different mutations including mostly frame-shifting gross deletions and duplications and rarely point mutations in DMD gene. ...

Implications of increased S100β and Tau5 proteins in dystrophic nerves of two mdx mouse models for Duchenne muscular dystrophy.

Krishnan VS, Aartsma-Rus A, Overzier M, Lutz C, Bogdanik L, Grounds MD.Mol Cell Neurosci. 2020 Mar 30:103484. doi: 10.1016/j.mcn.2020.103484. Online ahead of print.PMID: 32240725

This study investigates changes with respect to increasing protein levels in dystrophic nerves of two mdx mouse models of Duchenne muscular dystrophy (DMD). ...

Biomarkers for Duchenne muscular dystrophy: myonecrosis, inflammation and oxidative stress.

Grounds MD, Terrill JR, Al-Mshhdani BA, Duong MN, Radley-Crabb HG, Arthur PG. Dis Model Mech. 2020 Mar 2;13(2):dmm043638. doi: 10.1242/dmm.043638.PMID: 32224496 Free PMC article. Review.

Duchenne muscular dystrophy (DMD) is a lethal, X-linked disease that causes severe loss of muscle mass and function in young children. ...

The Effectiveness and Value of Deflazacort and Exon-Skipping Therapies for the Management of Duchenne Muscular Dystrophy.
Agboola F, Lin GA, Fluetsch N, Walton SM, Rind DM, Pearson SD. J Manag Care Spec Pharm. 2020 Apr;26(4):361-366. doi: 10.18553/jmcp.2020.26.4.361.PMID: 32223597 Free article.

FSHD Journal Articles

April 1 – April 30, 2020

Phenotypic Variability Among Patients With D4Z4 Reduced Allele Facioscapulohumeral Muscular Dystrophy.
Ruggiero L, Mele F, Manganelli F, Bruzzese D, Ricci G, Vercelli L, Govi M, Vallarola A, Tripodi S, Villa L, Di Muzio A, Scarlato M, Bucci E, Antonini G, Maggi L, Rodolico C, Tomelleri G, Filosto M, Previtali S, Angelini C, Berardinelli A, Pegoraro E, Moggio M, Mongini T, Siciliano G, Santoro L, Tupler R. JAMA Netw Open. 2020 May 1;3(5):e204040. doi: 10.1001/jamanetworkopen.2020.4040.PMID: 32356886
IMPORTANCE: Facioscapulohumeral muscular dystrophy (FSHD) is considered an autosomal dominant disorder, associated with the deletion of tandemly arrayed D4Z4 repetitive elements...

Facioscapulohumeral Muscular Dystrophy-a Tale of Heterogeneity and the Power of Clinical Assessments.
Nguyen K, Robin JD. JAMA Netw Open. 2020 May 1;3(5):e205004. doi: 10.1001/jamanetworkopen.2020.5004. PMID: 32356881 No abstract available.

A quantitative method to assess muscle edema using short TI inversion recovery MRI.
Dahlqvist JR, Salim R, Thomsen C, Vissing J. Sci Rep. 2020 Apr 29;10(1):7246. doi: 10.1038/s41598-020-64287-8. PMID: 32350361 Free article.
Muscle inflammation is an important component of disease pathophysiology in several muscular dystrophies. Hyperintensities on MRI sequences with short TI inversion recovery (STIR) reflect edema, or inflammation (STIR+). ...In this study, we developed a quantitative STIR method, and tested its ability to identify STIR+ lesions in healthy controls and patients with Facioscapulohumeral muscular dystrophy and compared the results with visual STIR evaluation and quantitative T2 relaxation time mapping. .

PAX7 target gene repression associates with FSHD progression and pathology over one year.
Banerji CRS. Hum Mol Genet. 2020 Apr 29:ddaa079. doi: 10.1093/hmg/ddaa079. Online ahead of print. PMID: 32347924
Facioscapulohumeral muscular dystrophy (FSHD) is a prevalent, inherited skeletal myopathy linked to hypomethylation of the D4Z4 macrosatellite at chromosome 4q35. ...

Facioscapulohumeral muscular dystrophy 1 patients participating in the UK FSHD registry can be subdivided into 4 patterns of self-reported symptoms.
Banerji CRS, Cammish P, Evangelista T, Zammit PS, Straub V, Marini-Bettolo C. Neuromuscul Disord. 2020 Mar 12:S0960-8966(20)30063-8. doi: 10.1016/j.nmd.2020.03.001. Online ahead of print. PMID: 32327287
Facioscapulohumeral muscular dystrophy (FSHD) is an autosomal dominant incurable skeletal muscle disease. FSHD1 constitutes 95% of cases and is linked to truncation of the D4Z4 macrosatellite at 4q35.

Outcomes of scapulothoracic fusion in facioscapulohumeral muscular dystrophy: A systematic review.
Kord D, Liu E, Horner NS, Athwal GS, Khan M, Alolabi B. Shoulder Elbow. 2020 Apr;12(2):75-90. doi: 10.1177/1758573219866195. Epub 2019 Aug 14. PMID: 32313557 Review.
BACKGROUND: Facioscapulohumeral muscular dystrophy (FSHD) is a rare condition associated with selective weakness of the muscles of the upper arm, face, and shoulder girdle, negatively affecting daily activities. ...

Interpretation of the Epigenetic Signature of Facioscapulohumeral Muscular Dystrophy in Light of Genotype-Phenotype Studies.
Nikolic A, Jones TI, Govi M, Mele F, Maranda L, Sera F, Ricci G, Ruggiero L, Vercelli L, Portaro S, Villa L, Fiorillo C, Maggi L, Santoro L, Antonini G, Filosto M, Moggio M, Angelini C, Pegoraro E, Berardinelli A, Maioli MA, D'Angelo G, Di Muzio A, Siciliano G, Tomelleri G, D'Esposito M, Della Ragione F, Brancaccio A, Piras R, Rodolico C, Mongini T, Magdinier F, Salsi V, Jones PL, Tupler R. Int J Mol Sci. 2020 Apr 10;21(7):2635. doi: 10.3390/ijms21072635. PMID: 32290091 Free PMC article.
Facioscapulohumeral muscular dystrophy (FSHD) is characterized by incomplete penetrance and intra-familial clinical variability. ...

Transgenic mice expressing tunable levels of DUX4 develop characteristics facioscapulohumeral muscular dystrophy-like pathophysiology ranging in severity.
Jones TI, Chew GL, Barraza-Flores P, Schreier S, Ramirez M, Wuebbles RD, Burkin DJ, Bradley RK, Jones PL. Skelet Muscle. 2020 Apr 11;10(1):8. doi: 10.1186/s13395-020-00227-4. PMID: 32278354 Free PMC article.

BACKGROUND: All types of facioscapulohumeral muscular dystrophy (FSHD) are caused by the aberrant activation of the somatically silent DUX4 gene, the expression of which initiates a cascade of cellular events ultimately leading to FSHD pathophysiology. ...

It's not all about muscle: fibroadipogenic progenitors contribute to facioscapulohumeral muscular dystrophy.

Serra C, Wagner KR. J Clin Invest. 2020 May 1;130(5):2186-2188. doi: 10.1172/JCI136133. PMID: 32250345 Free article.

Facioscapulohumeral muscular dystrophy (FSHD) results from expression of the full-length double homeobox 4 (DUX4-FL) retrogene in skeletal muscle. ...

Transcriptional and cytopathological hallmarks of FSHD in chronic DUX4-expressing mice.

Bosnakovski D, Shams AS, Yuan C, da Silva MT, Ener ET, Baumann CW, Lindsay AJ, Verma M, Asakura A, Lowe DA, Kyba M. J Clin Invest. 2020 May 1;130(5):2465-2477. doi: 10.1172/JCI133303. PMID: 32250341 Free article.

Facioscapulohumeral muscular dystrophy (FSHD) is caused by loss of repression of the DUX4 gene; however, the DUX4 protein is rare and difficult to detect in human muscle biopsies, and pathological mechanisms are obscure. ...

DUX4-expressing immortalised FSHD lymphoblastoid cells express genes elevated in FSHD muscle biopsies, correlating with the early stages of inflammation.

Banerji CRS, Macura MP, Zammit PS. Hum Mol Genet. 2020 Apr 2:ddaa053. doi: 10.1093/hmg/ddaa053. Online ahead of print. PMID: 32242220

Facioscapulohumeral muscular dystrophy (FSHD) is an incurable disorder linked to ectopic expression of DUX4. However, DUX4 is notoriously difficult to detect in FSHD muscle cells, while DUX4 target gene expression is an inconsistent biomarker for FSHD skeletal muscle biopsies, displaying efficacy only on pathologically inflamed samples. ...

Applying genome-wide CRISPR-Cas9 screens for therapeutic discovery in facioscapulohumeral muscular dystrophy.

Lek A, Zhang Y, Woodman KG, Huang S, DeSimone AM, Cohen J, Ho V, Conner J, Mead L, Kodani A, Pakula A, Sanjana N, King OD, Jones PL, Wagner KR, Lek M, Kunkel LM. Sci Transl Med. 2020 Mar 25;12(536):eaay0271. doi: 10.1126/scitranslmed.aay0271. PMID: 32213627

Here, we demonstrate the utility of a genome-wide CRISPR-Cas9 loss-of-function library to identify therapeutic targets for facioscapulohumeral muscular dystrophy (FSHD), a genetically complex type of muscular dystrophy for which there is currently no treatment. ...

A hospital based epidemiological study of genetically determined muscle disease in south western Norway.

Husebye SA, Rebne CB, Stokland AE, Sanaker PS, Bindoff LA. Neuromuscul Disord. 2020 Mar;30(3):181-185. doi: 10.1016/j.nmd.2020.01.006. Epub 2020 Feb 4. PMID: 32146000 Free article.
Myotonic dystrophy was the commonest adult muscle disorder with a minimum prevalence of 11.84/100,000 followed by facioscapulohumeral muscular dystrophy at 6.42/100,000. ...Becker muscular dystrophy was rare (0.4/100,000). For the purposes of comparison, we also ascertained adults with spinal muscular atrophy (SMA) and found a prevalence of 4.42/100,000. ...

Does DNA Methylation Matter in FSHD?

Salsi V, Magdinier F, Tupler R. Genes (Basel). 2020 Feb 28;11(3):258. doi: 10.3390/genes11030258. PMID: 32121044 Free PMC article. Review.
Facioscapulohumeral muscular dystrophy (FSHD) has been associated with the genetic and epigenetic molecular features of the CpG-rich D4Z4 repeat tandem array at 4q35. ...

Consequences of epigenetic derepression in facioscapulohumeral muscular dystrophy.

Greco A, Goossens R, van Engelen B, van der Maarel SM. Clin Genet. 2020 Feb 21. doi: 10.1111/cge.13726. Online ahead of print. PMID: 32086799 Review.
Facioscapulohumeral muscular dystrophy (FSHD), a common hereditary myopathy, is caused either by the contraction of the D4Z4 macrosatellite repeat at the distal end of chromosome 4q to a size of 1 to 10 repeat units (FSHD1) or by mutations in D4Z4 chromatin modifiers such as Structural Maintenance of Chromosomes Hinge Domain Containing 1 (FSHD2). ...

SMCHD1 promotes ATM-dependent DNA damage signaling and repair of uncapped telomeres.

Vančevska A, Ahmed W, Pfeiffer V, Feretzaki M, Boulton SJ, Lingner J. EMBO J. 2020 Apr 1;39(7):e102668. doi: 10.15252/embj.2019102668. Epub 2020 Feb 21. PMID: 32080884
Structural maintenance of chromosomes flexible hinge domain-containing protein 1 (SMCHD1) has been implicated in X-chromosome inactivation, imprinting, and DNA damage repair, and mutations in SMCHD1 can cause facioscapulohumeral muscular dystrophy. ...

Bilateral scapulothoracic arthrodesis for facioscapulohumeral muscular dystrophy: function, fusion, and respiratory consequences.

Boileau P, Pison A, Wilson A, van der Meijden O, Sacconi S, Trojani C, Gauci MO. J Shoulder Elbow Surg. 2020 May;29(5):931-940. doi: 10.1016/j.jse.2019.10.006. Epub 2020 Jan 22. PMID: 31982337
BACKGROUND: Scapulothoracic arthrodesis (STA) has been proposed for the treatment of painful scapular winging in patients with facioscapulohumeral muscular dystrophy (FSHD). ...

Magnetic resonance imaging correlates with electrical impedance myography in facioscapulohumeral muscular dystrophy.

Hamel J, Lee P, Glenn MD, Burka T, Choi IY, Friedman SD, Shaw DWW, McCalley A, Herbelin L, Dimachkie MM, Lemmers R, van der Maarel SM, Barohn RJ, Tawil R, Statland JM. Muscle Nerve. 2020 May;61(5):644-649. doi: 10.1002/mus.26792. Epub 2020 Jan 22. PMID: 31884698
INTRODUCTION: Electrical impedance myography (EIM) has been proposed as a noninvasive biomarker of muscle composition in facioscapulohumeral muscular dystrophy (FSHD). ...